# Testing Relations

## Graphs for three or more variables

An example study on disasters, food prices, and conflict

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# Today

1. The research question, and how the paper is built
2. Reading figures that show three or more variables
3. A follow-up paper that zooms in on one step
4. Making a three-variable graph yourself

**The one idea to take away**

Every variable in a graph needs its own visual **job**. Position, colour, size, shape, and panels are the jobs on offer.

# 01
# The example study

# The question

## Do floods and storms lead to more violent conflict?

### And does that depend on El Niño and La Niña?

- Floods and storms destroy harvests and disrupt supply chains
- That can push up food prices
- Food price shocks have been linked to unrest and fighting
- El Niño and La Niña shift where and how disasters hit

Your reading

D'Amico (2026). The price of instability. *Humanities and Social Sciences Communications*.

# How the paper is built

**1 Introduction**
The puzzle, and why it matters for policy

**2 Literature review**
What we know about climate, food, and conflict

**3 Theory**
Disaster → food prices → conflict, with a delay

**4 Data and methods**
Subnational regions, month by month, 2000 to 2021

**5 Results**
The figures carry the main findings

**6 Conclusion**
What it means, and what we still do not know

Today we spend most of our time on step 5.

# Where the data come from

| What | Source | Unit |
|---|---|---|
| Disasters | EM-DAT International Disaster Database | Event |
| Conflict | UCDP Georeferenced Event Dataset | Battle event |
| Food prices | World Food Programme (VAM) | Market price |
| El Niño / La Niña | NOAA Oceanic Niño Index | Month |

All four are matched to the same subnational regions and months. That merge is most of the work in a project like this.

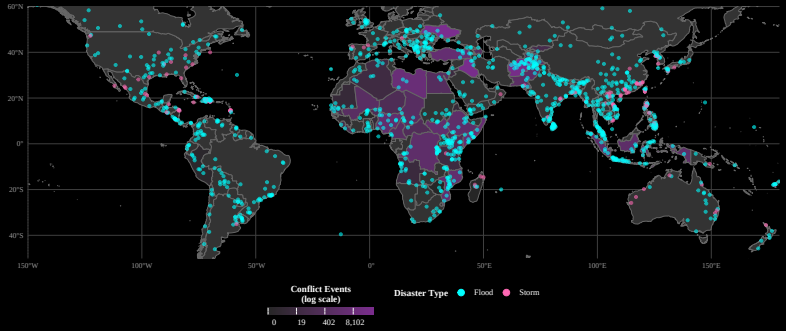
# 02
# Reading the figures

# Ways to show a variable

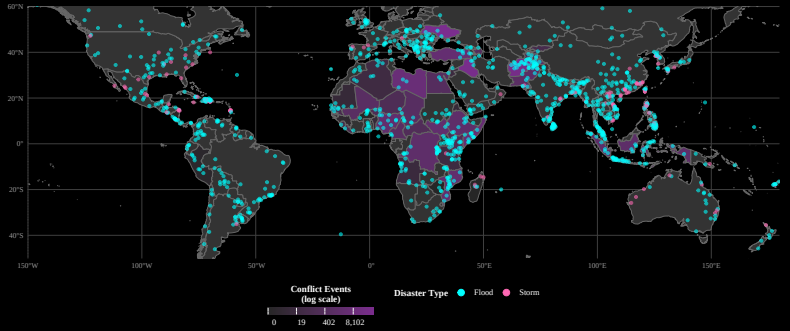
| Visual job | Good for | Example |
|---|---|---|
| Position on x or y | Numbers, time | Months since a flood |
| Colour | Groups | El Niño vs. La Niña |
| Size | A second number | Number of storms |
| Filled or open | Yes / no | Statistically significant or not |
| Panels | One more group | One panel per disaster type |

For each figure that follows, ask **what are the variables, and what job does each one have?**

# Where disasters and conflict overlap

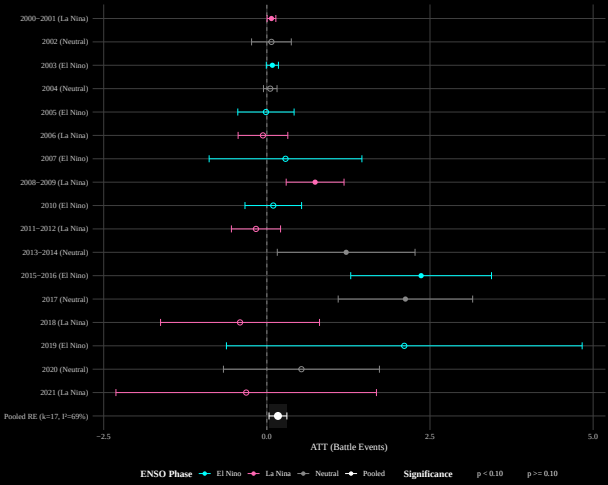


# Where disasters and conflict overlap

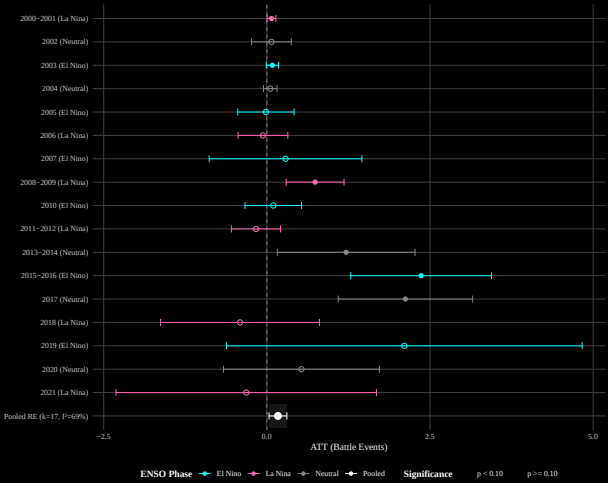



Location **map position** Disaster type **point colour** Conflict **country shading**

# Floods increase battle events



# Floods increase battle events




- **Rows** one El Niño or La Niña episode each
- **Dot** the estimated effect
- **Line** uncertainty around it
- **Colour** El Niño, La Niña, Neutral
- **Filled vs. open** significant or not

> The white dot at the bottom combines all 17 episodes.

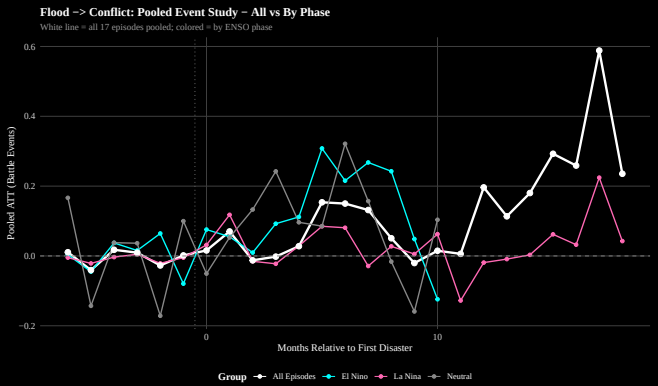
# Conflict peaks five to six months later




- **x** months before and after the disaster
- **y** effect on battle events
- Flat before the disaster, then a slow build

A delay like this fits a food price story better than people fleeing straight away.

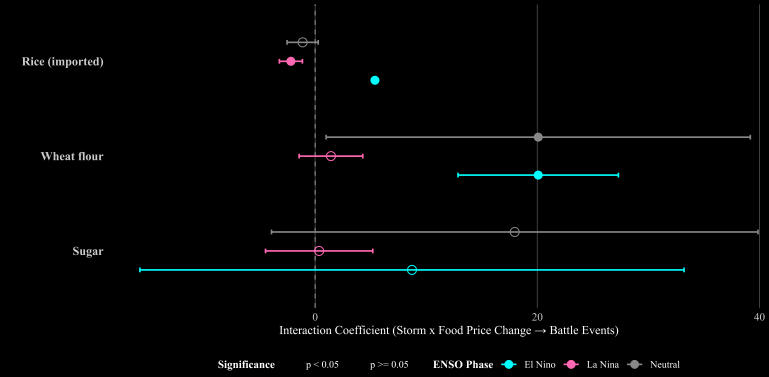
# The same figure, split by El Niño and La Niña



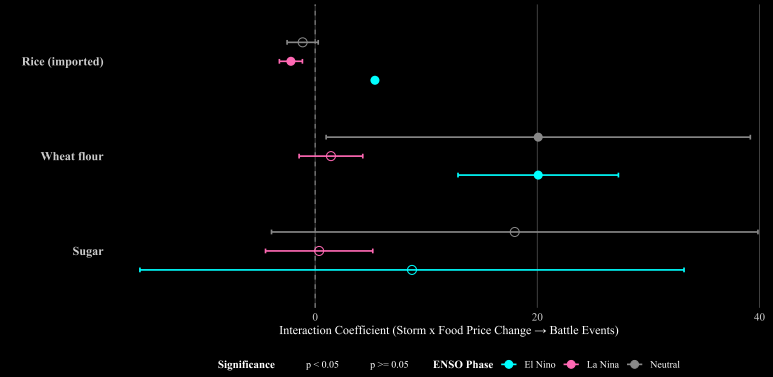

- **White** all 17 episodes, as on the last slide
- **Colour** one line per phase: El Niño, La Niña, Neutral

Adding colour as a third variable asks a new question. Does the delay show up in every phase?

# Four variables in one figure



# Four variables in one figure




Food type **rows** Effect **position** ENSO phase **colour** Significance **filled vs. open**

# Your turn 2 minutes

Think of a chart you have made, in Excel or anywhere else.

What third variable could you add?

And which visual job would you give it?

# 03
# A follow-up paper

# Zooming in on one step

Disaster → **food prices** → conflict

A second paper looks only at the middle step.

- Monthly wheat-flour prices in 129 regions of 19 countries
- Does the same disaster move prices the same way everywhere?
- Or does it depend on whether a region is in a **drier** or **wetter** spell?

Optional reading

D'Amico (2026). Regional El Niño patterns shape wheat-flour price responses to climate disasters. *Environmental Research Letters*.

# Same flood, opposite effects




- **x** months since the flood
- **y** change in wheat-flour price
- **Colour** drier vs. wetter regions
- **Band** uncertainty

Prices rise in drier regions and fall slightly in wetter ones. Without the colour, you would miss this.

# 04
# Make it yourself

# Start with a table

| Disaster | Drier | Wetter |
|---|---|---|
| Flood | 12.21 | −4.43 |
| Drought | 2.92 | 12.11 |
| Extreme temperature | 3.46 | 2.55 |
| Storm | −4.70 | 14.98 |

Average change in wheat-flour prices in the 13 months after each disaster

# Start with a table

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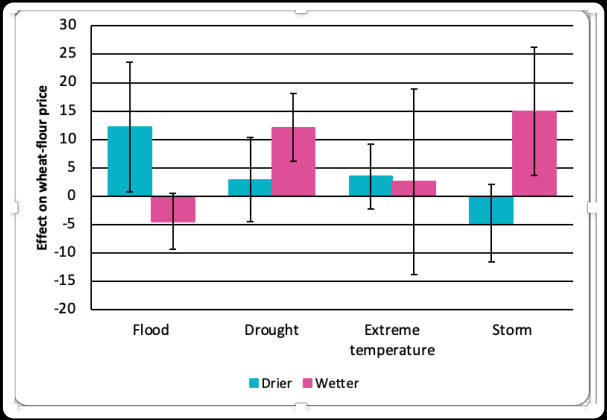
Average change in wheat-flour prices in the 13 months after each disaster

**Three variables** disaster type rows regional climate columns price effect the numbers

# In Excel

1. Select the table
2. **Insert → Clustered Column**
3. Click a bar, then **Add Chart Element → Error Bars**
4. Add axis titles

> Rows become groups on the x-axis. Columns become colours.



# Another one: trends over time

- **x** year, 2000 to 2021
- **Bar height** region-months with a flood
- **Colour** El Niño, La Niña, Neutral

The trick: put each year's number in the column for its phase. Excel then colours the bars for you.

Do El Niño years have more floods? Is that enough to say El Niño *causes* them?

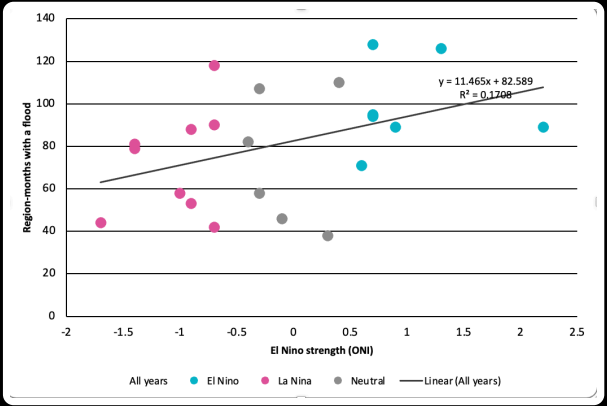
[Excel screenshot to add]

save it as slides/figures/excel_trend.png

# A scatter plot with a fitted line

- **x** how strong El Niño was that year
- **y** region-months with a flood
- **Colour** El Niño, La Niña, Neutral
- **Line** the best straight-line fit

The line slopes up, but the points are spread widely and there are only 22 years. How would you describe this in one honest sentence?



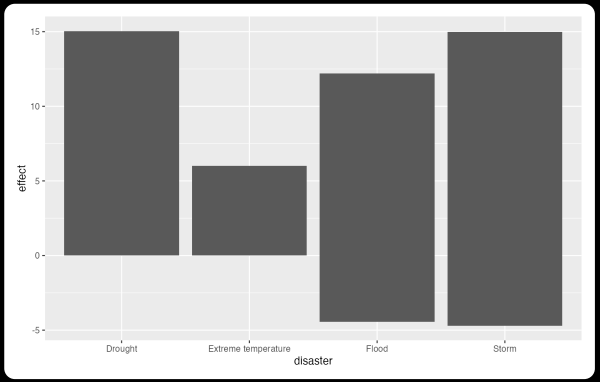
# The same chart in R

stage 1

```
d <- read.csv("pooled_effects.csv")

ggplot(d, aes(x = disaster,
              y = effect)) +
  geom_col()
```

Two variables only. Drier and wetter are piled on top of each other, so this chart is **wrong**.

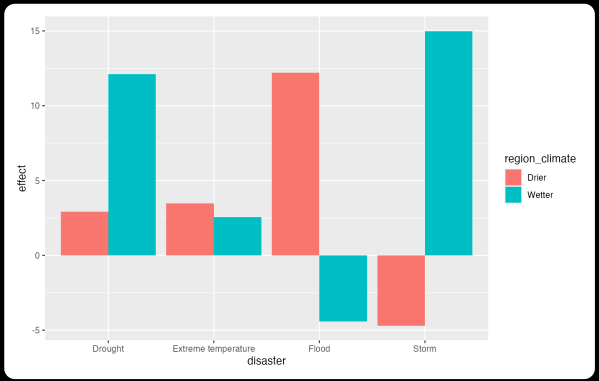


# The same chart in R

stage 2

```
ggplot(d, aes(x = disaster,
              y = effect,
              # NEW: third variable
              fill = region_climate)) +
  # NEW: side by side
  geom_col(position = "dodge")
```

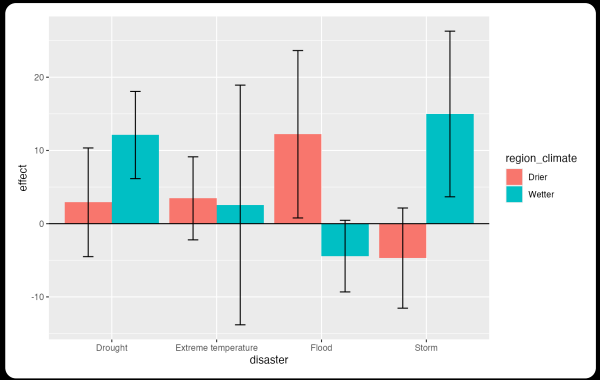
Colour gives the third variable its own job.



# The same chart in R

stage 3

```
ggplot(d, aes(x = disaster,
              y = effect,
              fill = region_climate)) +
  geom_col(position = "dodge") +
  # NEW: uncertainty
  geom_errorbar(
    aes(ymin = lower, ymax = upper),
    position = position_dodge(0.9),
    width = 0.2) +
  geom_hline(yintercept = 0)
```

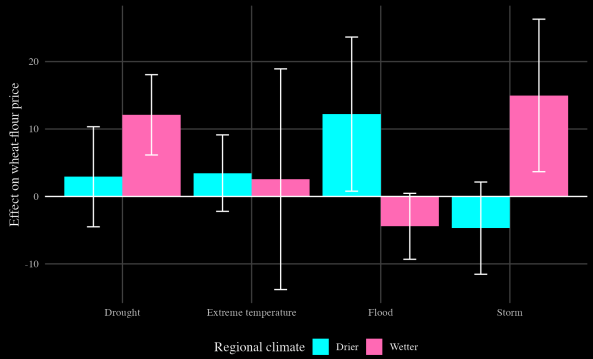



If an error bar crosses zero, we cannot rule out "no effect".

# The same chart in R

stage 4

```
... +
# NEW: colours and labels
scale_fill_manual(values = c(
  "Drier"  = "#00FFFF",
  "Wetter" = "#FF69B4")) +
labs(x = NULL,
     y = "Effect on wheat-flour price",
     fill = "Regional climate") +
# NEW: the dark look
theme_minimal() +
theme(plot.background =
        element_rect(fill = "black"),
      text =
        element_text(colour = "white"))
```



# A checklist for your own graphs

- Can you name every variable, and the job each one has?
- Does each variable have **one** job?
- Are the axes labelled in plain words, with units?
- Does the graph show uncertainty where there is some?
- Does the caption tell the reader what to look for?

**Start from the table**

If you can lay your data out as rows, columns, and numbers, you already have a three-variable chart.

# Play with an interactive visualization




**Bridging the Distance**

New insights on geography in conflict mediation

- A different study, on where peace talks happen
- The same research told as an interactive web page
- Click a country to explore the data

tvancisin.github.io/elisa_damico

Click the image or the link to open it

Web and visualizations by Tomas Vancisin

# Want to try it on your own?

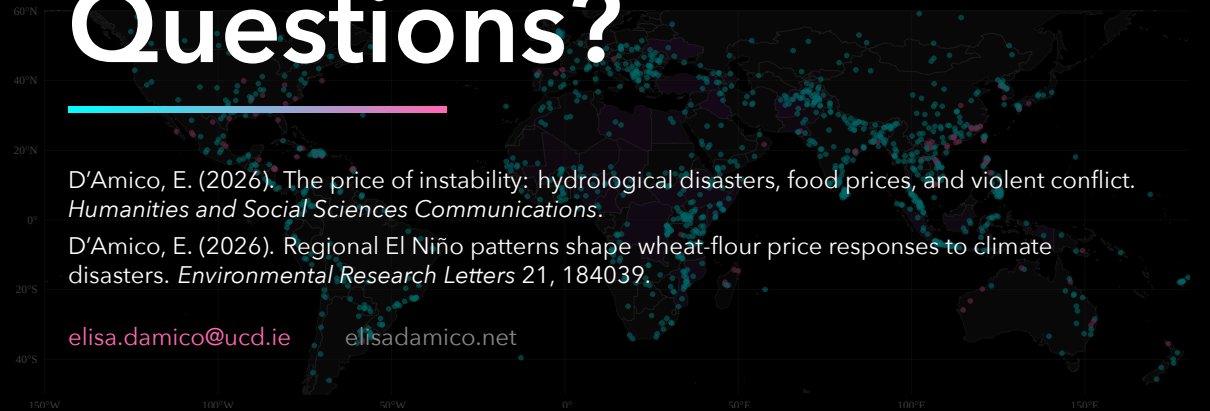
- Today's table on disasters and food prices
- Floods per year, 2000 to 2021, and El Niño strength
- Excel steps for three charts
- Today's R code, runs in your browser
- The interactive visualization

> Completely optional. Not part of your assignment.



`github.com/elisadamico/graphs-three-variables`

# Questions?



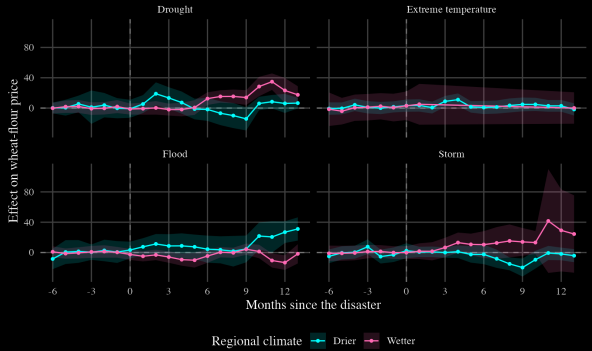

D'Amico, E. (2026). The price of instability: hydrological disasters, food prices, and violent conflict. *Humanities and Social Sciences Communications*.

D'Amico, E. (2026). Regional El Niño patterns shape wheat-flour price responses to climate disasters. *Environmental Research Letters* 21, 184039.

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# Appendix A a fourth variable with panels

```
ggplot(es, aes(x = month,
               y = effect,
               colour = region_climate,
               fill = region_climate)) +
  geom_ribbon(aes(ymin = lower,
                  ymax = upper),
              alpha = 0.15,
              colour = NA) +
  geom_line() +
  # NEW: fourth variable
  facet_wrap(~ disaster)
```



# Appendix B

## storms increase battle events too

